

Monotone Comparative Statics

To begin with a simple but important case, let X and T be nonempty subsets of $\mathbb{R}$ and let $f : X \times T \rightarrow \mathbb{R}$. The first question is: How does the set of maximizers of $f(x, t)$ on X vary with the parameter t ?

Definition. f satisfies the **strict single-crossing property (SSCP)** in (x, t) if for every x'', x' in X and t'', t' in T , with $x'' > x'$, and $t'' > t'$

$$f(x'', t') \geq f(x', t') \text{ implies } f(x'', t'') > f(x', t'').^1$$

The following simple Comparative Statics Theorem has many applications.

Theorem (Comparative Statics).² Let f satisfy the strict single-crossing property (SSCP) in (x, t) and let t_1, t_0 be any two points in T with $t_1 > t_0$. If x_i maximizes $f(x, t_i)$ on X for $i = 0, 1$, then $x_1 \geq x_0$.

Note that the only assumption we made on f is the SSCP. X and T can be finite or infinite; if they are intervals, f need not be concave or differentiable or even continuous in x .

Exercise 1. Prove the Comparative Statics Theorem. (Hint: just write down what it means for x_i to solve the problem for $t = t_i$. Then stare at the SSCP.)

Exercise 2. Suppose that $f(q, p) = pq - c(q)$, where $q \in \mathbb{R}_+$ and $p \in \mathbb{R}_{++}$. Show that f satisfies the strict single crossing property in (q, p) , without assuming that c is differentiable. By the Theorem, therefore, profit-maximizing supply of output is nondecreasing in the output price. (Although we know cost functions are increasing, note that the conclusion does not rely on that fact.)

Exercise 3. Suppose that X is an interval, that $f(\cdot, t)$ is continuously differentiable on X for every $t \in T$, and that $f_x(x, \cdot)$ is strictly increasing on T every $x \in X$. Prove that f satisfies the SSCP.³ (Hint: start by writing $f(x'', t) - f(x', t) = \int_{x'}^{x''} f_x(x, t) dx$.)

Exercise 4. ★ Suppose that X is an interval, that $f(\cdot, t)$ is continuously differentiable on X for every $t \in T$, and that for every t', t'' in T with $t'' > t'$ there is a positive number γ such that $f_x(x, t'') > \gamma f_x(x, t')$ for every $x \in X$. Prove that f satisfies the SSCP.

Exercise 5. ★ Consider MWG Exercise 3.G.4(b) for $L = 2$. Reformulate the problem so you can apply the Comparative Statics Theorem to conclude that the demand for each good is normal.

¹Note that the order matters: if f satisfies the SSCP in (x, t) , then it does not follow that it satisfies the SSCP in (t, x) .

²Easy Corollary to Milgrom and Shannon (1994), Theorem 4'.

³The conclusion remains true if you interchange the roles of t and x in this first sentence. Proof?

Increasing Differences and Supermodularity: Euclidean Spaces

In what follows X , Y , and T are *nonempty subsets of a Euclidean space* (not necessarily of the same dimension). Recall for vectors in $\mathbb{R}^n$, $y' \geq y$ means $y'_i \geq y_i$ for $i = 1, \dots, n$; and $y' > y$ means $y' \geq y$ but $y' \neq y$. Formally any Euclidean space is *partially ordered* by the usual inequality $>$. The definition of the **SSCP** here is *identical* to that on the first page, except that x and t can be vectors, rather than real numbers.

Definition. A function $f : X \times T \rightarrow \mathbb{R}$ satisfies **increasing differences** in x and t if for every x'', x' in X with $x'' > x'$, $f(x'', t) - f(x', t)$ is increasing in t . It satisfies **strictly increasing differences** if the difference is strictly increasing ($t'' > t'$ and $x'' > x'$ implies that $f(x'', t'') - f(x', t'') > f(x'', t') - f(x', t')$).

Fact. If f satisfies strictly increasing differences in x and t then f satisfies the strict single crossing property in (x, t) .

SUPERMODULARITY

For any $y, \tilde{y}$ in $\mathbb{R}^n$, let $y \vee \tilde{y} = (\max\{y_1, \tilde{y}_1\}, \dots, \max\{y_n, \tilde{y}_n\})$, the least upper bound of $\{y, \tilde{y}\}$ in $\mathbb{R}^n$; and $y \wedge \tilde{y} = (\min\{y_1, \tilde{y}_1\}, \dots, \min\{y_n, \tilde{y}_n\})$, the greatest lower bound of $\{y, \tilde{y}\}$ in $\mathbb{R}^n$.

Definition. A set Y is a **sublattice** (of $\mathbb{R}^n$) if for every y, y' in Y , $y \vee y' \in Y$ and $y' \wedge y \in Y$.

For example, a finite product of nonempty subsets of the real line is a sublattice. A budget set with positive prices and wealth is not. (What about a production set?)

Definition. Let Y be a sublattice. A function $f : Y \rightarrow \mathbb{R}$ is **supermodular** (SPM) if for every $\tilde{y}, y$ in Y

$$f(\tilde{y} \vee y) + f(\tilde{y} \wedge y) \geq f(\tilde{y}) + f(y).$$

If the inequality is strict whenever $\tilde{y} \not\geq y$ and $y \not\geq \tilde{y}$ then f is **strictly supermodular** (SSPM).

Facts. Let Y be a sublattice of $\mathbb{R}^n$ and $f : Y \rightarrow \mathbb{R}$.

1. f is [S]SPM if and only if f satisfies pairwise [S]ID: For every $i \neq j$, f satisfies [S]ID in (y_i, y_j) for every fixed level of the remaining variables.
2. Let Y be open (or the closure of an open set) and f be C^2 . (a) f is SPM if and only if $\partial^2 f / \partial y_i \partial y_j \geq 0$ on Y for every pair of variables i and j . (b) If $\partial^2 f / \partial y_i \partial y_j > 0$ on Y for every pair of variables i and j , then f is SSPM.
3. If f and g are [S]SPM, then so is $f + g$ and αf for $\alpha > 0$.

Theorem. (Topkis, 1978) Let X be a sublattice, let $f(\cdot, t)$ be supermodular on X for every $t \in T$, let f satisfy strictly increasing differences in (x, t) on $X \times T$, and let t^1, t^0 be any two points in T with $t^1 > t^0$. If x^i maximizes $f(x, t^i)$ on X for $i = 0, 1$, then $x^1 \geq x^0$.

Proof of the Theorem: Let f be SPM on X for each $t \in T$ and satisfy strictly increasing differences in (x, t) . Let t^1 and t^0 be two points in T with $t^1 > t^0$, let x^0 maximize $f(x, t^0)$ on X , and let x^1 be any element of X with $x^1 \not\geq x^0$. The last property implies that $x^0 \vee x^1 > x^1$. I will show that x^1

cannot maximize $f(\cdot, t^1)$ on X . Since x^0 maximizes $f(\cdot, t^0)$ on X and $x^0 \wedge x^1 \in X$, $f(x^0, t^0) \geq f(x^0 \wedge x^1, t^0)$. That inequality and **supermodularity** of f at t_0 implies $f(x^0 \vee x^1, t^0) \geq f(x^1, t^0)$. Finally the last inequality and **strictly increasing differences** implies $f(x^0 \vee x^1, t^1) > f(x^1, t^1)$. (Recall that $t^1 > t^0$, and $x^0 \vee x^1 > x^1$.) Since $x^0 \vee x^1 \in X$, x^1 does not maximize $f(\cdot, t^1)$ on T . Equivalently, if x^1 maximizes $f(\cdot, t^1)$ on T , then $x^1 \geq x^0$. $\square$

Corollary. *Suppose that f is C^2 on $Y = X \times T$. If $\partial^2 f / \partial y_i \partial y_j > 0$ on Y for every pair of variables i and j (whether they are part of x or t), then the conclusion of Topkis's Theorem holds.*

Single-Crossing Property for Euclidean Spaces

Let X and T and Y each be the nonempty subsets of some Euclidean space, as in the last section. The definition of the strict single crossing property is exactly the same as before. The next definition generalizes the notion of supermodularity.

Definition. *Let Y be a sublattice. A function $f : Y \rightarrow \mathbb{R}$ is quasisupermodular if for every $\tilde{y}$ and y in Y*

$$f(y) \geq [>]f(y \wedge \tilde{y}) \quad \text{implies} \quad f(y \vee \tilde{y}) \geq [>]f(\tilde{y}).$$

It is easy to show that quasisupermodularity is preserved under increasing transformations; in particular an increasing transformation of a supermodular function is quasisupermodular. We now have a crowning result. The only differences between it and Topkis's Theorem are the words in bold. In what follows $f : X \times T \rightarrow \mathbb{R}$.

Theorem (Milgrom and Shannon, 1994, Theorem 4'). *Let X be a sublattice, $f(\cdot, t)$ be **quasisupermodular** for every $t \in T$, let f satisfy the **strict single crossing property** in (x, t) on $X \times T$, and let t^1, t^0 be any two points in T with $t^1 > t^0$. If x^i maximizes $f(x, t^i)$ on X for $i = 0, 1$, then $x^1 \geq x^0$.*

The proof is *identical* to my proof of Topkis's Theorem *except that* “quasisupermodular” replaces “supermodular” and “strict single crossing property” replaces “strictly increasing differences.” (I've underlined and put in **red** the words which change.)

Extension Beyond Euclidean Spaces

The results presented so far are for Euclidean spaces endowed with the usual relation “ $\geq$ ” between vectors in Euclidean spaces. The results can be extended beyond this important special case.

First some definitions. A *partially-ordered set* Y is a set Y and a binary relation $R \subset Y \times Y$ that is *reflexive* (yRy), *transitive*, and *antisymmetric* (xRy and yRx implies $x = y$). Note that “ $\geq$ ” on a Euclidean space is a partial order. For a partially ordered set Y , define $y \wedge y'$ to be the greatest lower bound of y and y' and $y \vee y'$ the least upper bound. As in the case of Euclidean space, a partially-ordered set Y is a *sublattice* if for every y, y' in Y , $y \vee y' \in Y$ and $y \wedge y' \in Y$.

Now let X be a sublattice with partial order R and T a partially ordered set with partial order $\succsim$. Let P be the strict part of R and $\succ$ the strict part of $\succsim$. A function $f : X \times T \rightarrow \mathbb{R}$ satisfies the *strict single-crossing property* if for every x, x' in X with $x'Px$, t, t' in T with $t' \succ t$, $f(x', t) - f(x, t) \geq 0$ implies $f(x', t') - f(x, t') > 0$. The definition of (weak) quasisupermodularity requires no modification (given that the definition of the meet and join depend on the partial order).

The Euclidean version of Milgrom-Shannon's Theorem 4' requires just a minor modification for this more general setting.

Theorem (Milgrom and Shannon, 1994, Theorem 4'). *Let X be a sublattice, $f(\cdot, t)$ be (weakly) **quasisupermodular** on X for every $t \in T$, let f satisfy the **strict single crossing property** in (x, t) on $X \times T$, and let t^1, t^0 be any two points in T with $t^1 \succ t^0$. If x^i maximizes $f(x, t^i)$ on X for $i = 0, 1$, then $x^1 Rx^0$.*

Applications

I recommend Milgrom and Shannon (1994), pp. 167-174, for several applications. I also have several problems on ASMT V-a (not to be turned in) that I reproduce here. I will publish notes on these problems to Canvas before the midterm.

1. A firm produces 2 outputs $q = (q_1, q_2)$ that it sells at prices $(p_1, p_2) \gg 0$ according to a technology represented by a cost function $c(q_1, q_2, t)$, where t is a parameter describing the state of the technology.⁴ Suppose that $c(q_1, q_2, t) = -tg(q_1, q_2)$, where $t < 0$, $g > 0$, and g is C^1 with positive first derivatives. What additional condition(s) implies that the solutions to the firm's profit maximization problem are increasing in t in the sense that if q^i solves the firm's problem at $t = t_i$ with $0 > t_1 > t_0$, then $q^1 \geq q^0$? Interpret the condition(s). Can you think of a hypothetical example of a multiproduct firm whose cost function plausibly satisfies the condition(s)?
2. (From the 2010 midterm) A firm owns a tree and must decide at what date $x \geq 0$ to cut it down and sell the wood. If it cuts and sells the tree at date x its date- x profit is $g(x)$, where $g(x) > 0$ for all $x \geq 0$. The instantaneous interest rate is r , and the firm's objective is to choose the date $x \geq 0$ to maximize discounted profit,

$$g(x)e^{-rx}.$$

Let $0 < r_1 < r_0$ and suppose that for $i = 0, 1$, the date x_i solves the firm's problem when $r = r_i$. Can you say either that $x_1 \geq x_0$ or $x_0 \geq x_1$? Justify your answer. (Note: all you know about g is that it is a positive function.)

3. If a single-output technology exhibits constant returns then the cost function is of the form $c(q, w) = qg(w)$, for some function g . *Evaluate*: "If the technology in is represented by a strictly quasiconcave, monotone production function then no input is inferior in the sense that if $q_1 > q_0 > 0$ and z^t minimizes cost at q_t , then $z^1 > z^0$." Does $z^1 \geq z^0$ still hold if quasiconvavity replaces strict quasiconcavity? (Note: you may use the fact that the cost function is differentiable in w if the production function is strictly quasiconcave.)
4. A competitive firm produces a single output using a single input according to a differentiable but *not* necessarily concave production function f_t , where $t \in \{0, 1\}$ is parameter describing technological change. In particular $f'_1(z) > f'_0(z) > 0$ for all $z \geq 0$ and $f_t(0) = 0$ for $t = 0, 1$. The output price is $p > 0$ and the input price is 1.
 - (a) Let z_t maximize the firm's profit at price p when the technology is t . Does it follow that $z_1 \geq z_0$?
 - (b) Put this technology in the context of the two-good quasilinear economy with a single firm and single consumer; in particular the consumer is endowed with $\omega > 0$ units of good 2 (the input) and owns the firm, and has strongly monotone preferences which are quasilinear in good 2 on the consumption set $X = \mathbb{R}_+ \times \mathbb{R}$. Which economy does the consumer prefer, the one with technology 0 or technology 1 (if the allocation is determined by competitive equilibrium)?⁵

⁴How would you define the cost function for a two-output technology?

⁵Our proof of the First Welfare Theorem assumed that the production function was concave, preferences convex, so if you want to appeal to that Theorem, you'll need to give a new proof of it here.